

Liyakhla Ntuli

Exercise 01 -

Fundamentals: SELECT and filtering

Q1. Retrieve all employees

SELECT

FROM employees ;

ID	first - name	last - name	department	salary	hire - date	city
1	Alice	Green	IT	25000	2020-01-15	Johannesburg
2	Bruce	Lee	HR	15000	2019-05-10	Cape Town
3	Chloe	Zulu	Finance	30000	2018-03-01	Durban
4	David	Makorena	Marketing	20000	2021-10-01	Pretoria
5	Eva	Musodo	IT	22000	2017-04-20	Johannesburg

Q2. Find all unique departments

SELECT DISTINCT department

FROM employees ;

department
IT
HR
Finance
Marketing

Q3 - Names ordered by ~~id~~ descending

```
SELECT first_name, last_name  
FROM employees  
ORDER BY Salary DESC;
```

first_name	last_name
Eva	Nawlos
Alic	Green
Lally	Zulu
David	Molocoona
Brian	Lee

Q4 - Top 3 highest-paid employees

```
SELECT id, first_name, last_name, Salary  
FROM employees  
ORDER BY Salary  
LIMIT 3;
```

id	first_name	last_name	Salary
1	Eva	Nawlos	72 000
2	Alic	Green	70 000
3	Lally	Zulu	65 000

Q5 - Employees in IT department

```
SELECT id, first_name, last_name, department
FROM employee
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6 - Finance employees with salary > 60 000

```
SELECT id, first_name, last_name, department, salary
FROM employee
WHERE department = 'Finance'
AND salary > 60 000;
```

id	first_name	last_name	department	salary
	Cathy	Zulu	Finance	65 000

Q7 - Employees in HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employee
WHERE department = 'HR'
OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q 8 - Employees not in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE NOT department = 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q 9 - IT, HR, or Finance Using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q 10 - IT, Salary > 65 000, city Johannesburg

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT'
AND salary > 65 000
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
5	Eva	Wardoo	IT	71 000	Johannesburg